

Many books and tutorials represent the command line prompt as just `$` to simplify their examples. It's possible to customize these prompts, but doing so is beyond the scope of this book.

Running Programs from the Command Line

To run a program or command, enter its name into the command line. Let's run the default calculator program that comes with the operating system. Enter the following into the command line:

- On Windows, enter `calc.exe`.
- On macOS, enter `open -a Calculator`. (Technically, this runs the `open` program, which then runs the Calculator program.)
- On Linux, enter `gnome-calculator`.

Program names and commands are case sensitive on Linux but case insensitive on Windows and macOS. This means that even though you must type `gnome-calculator` on Linux, you could type `calc.exe` on Windows and `OPEN -a Calculator` on macOS.

Entering these calculator program names into the command line is equivalent to running the Calculator program from the Start menu, Finder, or Dash. These calculator program names work as commands because the `calc.exe`, `open`, and `gnome-calculator` programs exist in folders that are included in the `PATH` environment variables. “Environment Variables and `PATH`” on page 35 explains this further. But suffice it to say that when you enter a program name on the command line, the shell checks whether a program with that name exists in one of the folders listed in `PATH`. On Windows, the shell looks for the program in the `cwd` (which you can see in the prompt) before checking the folders in `PATH`. To tell the command line on macOS and Linux to first check the `cwd`, you must enter `./` before the filename.

If the program isn't in a folder listed in `PATH`, you have two options:

- Use the `cd` command to change the `cwd` to the folder that contains the program, and then enter the program name. For example, you could enter the following two commands: